Combinatorics of the critical loci of Deep Linear Networks

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Abstract

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1 Introduction

Flow of the paper: The main theorem will be the stratification of the critical loci via the stratification of the cyclic variety and its intersection wit the specialization. We introduce notation, we state the main theorem (without any quiver representation theory simply in terms of the rank patterns).

2 Related works

3 Setup and main theorem

The central result of this paper is that the algebraic variety of critical points of the loss landscape of a deep linear network admits a stratification parametrized by a finite set of geometric invariants which are given by some rank patterns. These rank patterns are the rank of a product made of weight matrices and some projection of the input-output covariance matrix.

Definition 3.1 (Deep Linear Network)**.** A L-layers Deep Linear Network (DLN) is a function:

$$f : \mathbb{R}^{d_0} \times \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}} \to \mathbb{R}^{d_L}$$
$$(x, \theta = (W_1, ..., W_L)) \mapsto \mu(\theta)x = W_L ... W_1 x$$

The dimension d_0 is the dimension of the input data and the dimensions d_l, $1 \leq l \leq L$ are the width of the layers. A Deep Linear Network is a multi-linear function (in parameter space $\Omega = \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}}$) which is linear in each of the weight matrix W_l and the data. It is linear in function space $\mathcal{F} := \mathrm{Mat}_{d_L \times d_0}$ over $\mu(\theta) := W := W_L ... W_1$ where μ is the product map. For the input-output data $(X, Y) \sim P$ we have the input and input-out covariance matrices $\Sigma_X = \mathbb{E}[XX^\top]$ and $\Sigma_{YX} = \mathbb{E}[YX^\top]$ and we will work with the quadratic population loss:

$$L(\theta) := ||M - \mu(\theta)\Sigma_x^{-\frac{1}{2}}||_F^2$$

where $||\cdot||_F$ is the Frobenius norm and $M := \Sigma_{YX}\Sigma_X^{-1}$ is the teacher matrix which is the ordinary least square regressor.

We define some algebraic varieties that will be important to derive the main result. The cyclic variety Γ_d is defined by a set of vanishing polynomial cyclic conditions which relates to the critical loci. This variety will be then stratified and the critical loci will inherit a stratification.

Definition 3.2 (Determinantal, cyclic, and critical loci)**.** Define

$$\Sigma_d^r := \{\theta \in \Omega : \mathrm{rank}(\mu(\theta)) = r\} \quad \textit{Determinantal variety},$$

$$\Gamma_d := \{(\theta, W_0) \in \Omega \times \mathrm{Mat}_{d_0 \times d_L} \mid W_{<\ell} W_0 W_{>\ell} = 0 \text{ for all } 1 \leq \ell \leq L\}, \quad \textit{Cyclic variety}$$

Let $\mathcal{F}^t := \mathrm{Mat}_{d_0 \times d_L}$ and let $\pi : \Gamma_d \to \mathcal{F} \times \mathcal{F}^t$ such that $\pi(\theta, W_0) := (\mu(\theta), W_0)$. Define the determinantal cyclic variety:

$$\Gamma_d^r := \Gamma_d \cap (\Sigma_d^r \times \mathrm{Mat}_{d_0 \times d_L})$$

Let S be a subset of left-singular vectors of M, define the S-critical loci:

$$\mathcal{C}_d^S = \{\theta \in \Omega | \ \nabla_\theta L(\theta) = 0; \mu(\theta) = P_S W^\star\}$$

In proposition **??** we show that the S- critical loci is a fiber of π:

$$\mathcal{C}_d^S \cong \Gamma_d^S := \pi^{-1}(P_S W^\star, \Sigma_{XY} P_Q) \subset \Gamma_d^r$$

Definition 3.3 (Symmetry group)**.** For dimension vector $d = (d_0, ..., d_L)$ define the symmetry group:

$$G_d = \prod_{i=0}^{L} GL_{d_i}$$

Consider the action of G_d:

$$g.(\theta, W_0) = (g_L W_L g_{L-1}^{-1}, \ldots, g_1 W_1 g_0^{-1}, g_0 W_0 g_L^{-1})$$

and note that we have a well defined action of G_d over the cyclic variety from G_d-invariance of the equations defining the cyclic variety. The variety Γ_d^r also inherits an action of G_d from the action of G_d over $\Sigma_d^r \times \mathrm{Mat}_{d_0 \times d_L}$, since the rank is invariant under action of GL_d. A consequence of the action is that we can stratify the variety Γ_d^r into orbits (strata) O_ρ. In this paper we will show that the relevant geometric invariant ρ parametrizing the orbits are the cyclic rank patterns.

Definition 3.4. For $\phi = (\theta, W_0) \in \Gamma_d$ we define its *cyclic rank pattern* to be the array

$$\rho(\phi) = (r_{ij}(\phi))_{0 \le i,j \le L} \quad \text{where} \quad \rho_{ij}(\phi) := \mathrm{rank}\big(W_j ... W_i\big) \quad \text{for} \quad \text{and} \quad \rho_{ii}(\phi) = d_i \text{and} \quad \rho_{i+1i}(\phi) = \mathrm{rank}\big(W_i\big)$$

Theorem 3.5. Define the strata $\mathcal{O}_\rho^S := \mathcal{O}_\rho \cap \Gamma_d^S$ where $\mathcal{O}_\rho$ is a G_d-orbit of the cyclic variety Γ_d^r. Assuming that the data is non degenerate and the input covariance matrix is invertible then the loci of critical points of rank $r < r_{\max}$ admits the following stratification parameterized by subset S of singular vectors of the teacher with r elements and admissible cyclic rank patterns $\rho \in R_d^{\mathrm{orb}}$:

$$\mathcal{C}_d^r = \bigsqcup_{|S|=r} \mathcal{C}_d^S = \bigsqcup_{|S|=r} \bigsqcup_{\substack{\rho \in R_d^{\mathrm{orb}} \\ \mathrm{rank}(\rho) = \mathrm{rank}(\Sigma_{XY} P_Q)}} \mathcal{O}_\rho^S$$

4 A stratification of the cyclic variety induces a stratification of the critical loci

In this section we explain how the orbit stratification of the cyclic variety induces a genuine orbit stratification of each fixed-S critical locus. All closures are relative Zariski closures.

Fix a subset S of left singular vectors and set

$$B_S := P_S W^\star, \qquad E_S := \Sigma_{XY} P_Q.$$

Thus $B_S \in \mathcal{F} = \mathrm{Mat}_{d_L \times d_0}$ and $E_S \in \mathcal{F}^t = \mathrm{Mat}_{d_0 \times d_L}$. Recall that

$$\pi : \Gamma_d \longrightarrow \mathcal{F} \times \mathcal{F}^t, \qquad \pi(\theta, W_0) := (\mu(\theta), W_0).$$

We write

$$\Gamma_d^S := \pi^{-1}(B_S, E_S) = \{(\theta, W_0) \in \Gamma_d \mid \mu(\theta) = B_S,\ W_0 = E_S\}.$$

If $\mathrm{rank}(B_S) = r$, then $\Gamma_d^S \subseteq \Gamma_d^r$.

Proposition 4.1 (The S-critical locus as a fiber)**.** Assume that Σ_X is invertible and that

$$W^\star = \Sigma_{YX} \Sigma_X^{-1}.$$

Then the map

$$\Theta_S : \mathcal{C}_d^S \longrightarrow \Gamma_d^S, \qquad \Theta_S(\theta) := (\theta, E_S),$$

is an isomorphism of varieties, with inverse given by projection onto the θ-coordinates. In particular,

$$\mathcal{C}_d^S \cong \pi^{-1}(P_S W^\star, \Sigma_{XY} P_Q).$$

Proof. Let $\theta \in \Omega$ satisfy $\mu(\theta) = B_S = P_S W^\star$. Then

$$\Sigma_X (P_S W^\star)^\top - \Sigma_{XY} = -\Sigma_{XY} P_Q = -E_S.$$

Hence, for every $1 \le \ell \le L$,

$$\left(\nabla_{W_\ell} \mathcal{L}(\theta)\right)^\top = -2 W_{<\ell} E_S W_{>\ell}.$$

Therefore the gradient equations are equivalent, under the constraint $\mu(\theta) = B_S$, to the cyclic equations

$$W_{<\ell} E_S W_{>\ell} = 0 \qquad \text{for every } 1 \le \ell \le L.$$

These are precisely the equations saying that

$$(\theta, E_S) \in \Gamma_d$$

together with $\pi(\theta, E_S) = (B_S, E_S)$. Thus

$$\theta \in \mathcal{C}_d^S \quad \Longleftrightarrow \quad (\theta, E_S) \in \Gamma_d^S.$$

The maps $\theta \mapsto (\theta, E_S)$ and $(\theta, E_S) \mapsto \theta$ are regular and inverse to one another. □

We now use the symmetry action. The group

$$G_d = \prod_{i=0}^{L} GL_{d_i}$$

acts on Γ_d by

$$g \cdot (\theta, W_0) = (g_L W_L g_{L-1}^{-1}, \ldots, g_1 W_1 g_0^{-1}, g_0 W_0 g_L^{-1}).$$

The map π is equivariant in the following sense:

$$\pi(g \cdot (\theta, W_0)) = (g_L \mu(\theta) g_0^{-1},\ g_0 W_0 g_L^{-1}).$$

Define the stabilizer of the fixed pair (B_S, E_S) by

$$K_S := \left\{ g \in G_d \;\middle|\; g_L B_S g_0^{-1} = B_S,\ g_0 E_S g_L^{-1} = E_S \right\}.$$

Then K_S acts on the fiber Γ_d^S.

Assume from now on that the rank-r cyclic variety has a finite G_d-orbit stratification by cyclic rank patterns:

$$\Gamma_d^r = \bigsqcup_{\rho \in R_d^r} O_\rho,$$

where R_d^r denotes the finite set of cyclic rank patterns occurring in Γ_d^r. We assume moreover[1] that the ambient cyclic orbit closures are governed by the coordinatewise rank order:

$$\overline{O_\rho}^{\Gamma_d^r} = \bigsqcup_{\substack{\rho' \in R_d^r \\ \rho' \le_{\mathrm{rk}} \rho}} O_{\rho'}.$$

[1]Both of these assumptions will be proven later on in the paper

Here

$$\rho' \leq_{\mathrm{rk}} \rho \quad \Longleftrightarrow \quad \rho'_{ij} \leq \rho_{ij} \quad \text{for every cyclic rank coordinate } (i,j).$$

For each $\rho \in R_d^r$, define

$$\mathcal{O}_\rho^S := O_\rho \cap \Gamma_d^S = O_\rho \cap \pi^{-1}(B_S, E_S).$$

Let

$$R_d^S := \{\rho \in R_d^r \mid \mathcal{O}_\rho^S \neq \varnothing\}.$$

Proposition 4.2 (Induced K_S-orbit stratification of the S-critical locus)**.** For every $\rho \in R_d^S$, the stratum

$$\mathcal{O}_\rho^S = O_\rho \cap \pi^{-1}(B_S, E_S)$$

is a single K_S-orbit. Consequently,

$$\Gamma_d^S = \bigsqcup_{\rho \in R_d^S} \mathcal{O}_\rho^S$$

is a finite decomposition into smooth locally closed K_S-orbits. Via the identification $\mathcal{C}_d^S \cong \Gamma_d^S$ from Proposition 4.1, this gives a stratification of $\mathcal{C}_d^S$.

Define the fixed-S closure relation on R_d^S by

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \subseteq \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}.$$

Equivalently, since the strata are K_S-orbits,

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \cap \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \neq \varnothing.$$

Then the closure of each stratum is the union of the strata below it for this relation:

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} = \bigsqcup_{\substack{\rho' \in R_d^S \\ \rho' \leq_S \rho}} \mathcal{O}_{\rho'}^S.$$

Equivalently, if

$$\rho' <_S \rho \quad \Longleftrightarrow \quad \rho' \leq_S \rho \text{ and } \rho' \neq \rho,$$

then the frontier satisfies

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \setminus \mathcal{O}_\rho^S = \bigsqcup_{\substack{\rho' \in R_d^S \\ \rho' <_S \rho}} \mathcal{O}_{\rho'}^S.$$

Moreover, the fixed-S closure order refines the ambient coordinatewise rank order:

$$\rho' \leq_S \rho \quad \Longrightarrow \quad \rho' \leq_{\mathrm{rk}} \rho.$$

Thus

$$\{\rho' \in R_d^S \mid \rho' \leq_S \rho\} \subseteq \{\rho' \in R_d^S \mid \rho' \leq_{\mathrm{rk}} \rho\}.$$

The inclusion may be strict.

Proof. First we show that each nonempty $\mathcal{O}_\rho^S$ is a single K_S-orbit. Let

$$x, y \in \mathcal{O}_\rho^S.$$

Since x and y both lie in the G_d-orbit O_ρ, there exists $g \in G_d$ such that

$$y = g \cdot x.$$

Because both x and y lie in the fiber Γ_d^S, we have

$$\pi(x) = (B_S, E_S), \qquad \pi(y) = (B_S, E_S).$$

Using the equivariance of π,

$$(B_S, E_S) = \pi(y) = \pi(g \cdot x) = (g_L B_S g_0^{-1},\, g_0 E_S g_L^{-1}).$$

Hence $g \in K_S$. Therefore $y \in K_S \cdot x$. Conversely, K_S preserves both the G_d-orbit O_ρ and the fiber Γ_d^S, so

$$K_S \cdot x \subseteq O_\rho \cap \Gamma_d^S.$$

Thus

$$\mathcal{O}_\rho^S = K_S \cdot x.$$

Since algebraic group orbits are smooth and locally closed, each $\mathcal{O}_\rho^S$ is smooth and locally closed. The decomposition

$$\Gamma_d^S = \bigsqcup_{\rho \in R_d^S} \mathcal{O}_\rho^S$$

follows by intersecting the disjoint orbit decomposition of Γ_d^r with the fiber Γ_d^S and discarding the empty intersections.

We now prove the frontier condition. Fix $\rho \in R_d^S$ and set

$$Z_\rho := \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}.$$

Because $\mathcal{O}_\rho^S$ is K_S-stable, its closure Z_ρ is also K_S-stable. Suppose

$$Z_\rho \cap \mathcal{O}_{\rho'}^S \neq \varnothing.$$

Choose $x \in Z_\rho \cap \mathcal{O}_{\rho'}^S$. Since $\mathcal{O}_{\rho'}^S$ is a K_S-orbit,

$$\mathcal{O}_{\rho'}^S = K_S \cdot x.$$

Since Z_ρ is K_S-stable,

$$K_S \cdot x \subseteq Z_\rho.$$

Therefore

$$\mathcal{O}_{\rho'}^S \subseteq Z_\rho.$$

This proves the frontier condition. The closure equality follows because the $\mathcal{O}_{\rho'}^S$'s form a finite disjoint decomposition of Γ_d^S.

It remains to compare $\leq_S$ with the coordinatewise rank order. Since

$$\mathcal{O}_\rho^S \subseteq O_\rho \qquad \text{and} \qquad \Gamma_d^S \subseteq \Gamma_d^r,$$

we have

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \subseteq \Gamma_d^S \cap \overline{O_\rho}^{\Gamma_d^r}.$$

Using the ambient cyclic closure formula,

$$\overline{O_\rho}^{\Gamma_d^r} = \bigsqcup_{\substack{\rho'' \in R_d^r \\ \rho'' \leq_{\mathrm{rk}} \rho}} O_{\rho''},$$

we obtain

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \subseteq \Gamma_d^S \cap \bigsqcup_{\substack{\rho'' \in R_d^r \\ \rho'' \leq_{\mathrm{rk}} \rho}} O_{\rho''} = \bigsqcup_{\substack{\rho'' \in R_d^S \\ \rho'' \leq_{\mathrm{rk}} \rho}} \mathcal{O}_{\rho''}^S.$$

Therefore, if $\rho' \leq_S \rho$, then

$$\mathcal{O}_{\rho'}^S \subseteq \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}$$

forces

$$\rho' \leq_{\mathrm{rk}} \rho.$$

This proves

$$\leq_S \subseteq \leq_{\mathrm{rk}}$$

on the set R_d^S. □

Remark 4.3. The relation $\leq_S$ is the actual frontier relation on the fixed-S critical locus. The coordinatewise order $\leq_{\mathrm{rk}}$ is only an ambient necessary condition for specialization. Thus one should not replace $\leq_S$ by $\leq_{\mathrm{rk}}$ in the closure formula.

Indeed, even if

$$\rho' \leq_{\mathrm{rk}} \rho,$$

the intersection

$$O_{\rho'} \cap \pi^{-1}(B_S, E_S)$$

may be empty. In that case there is no K_S-orbit $\mathcal{O}_{\rho'}^S$ in the fixed-S critical locus.

In the following sections we will show how to stratify Γ_d^r using quiver representation theory.

5 The cyclic rank patterns as geometric invariants of deep linear networks

Goal: Stratify the cyclic variety with cyclic rank patterns.

5.1 Essentials of Quiver representation theory

A quiver is a graph (I, E) with vertices I, edges E and an orientation. A quiver representation assigns a vector space to $V_i \cong \mathbb{R}^{d_i}$ and a linear map $\phi : V_i \to V_{i+1}$ with vector dimension d_i.

5.2 The cyclic Nakayama algebra

Throughout this subsection, put $n = L+1$, and take all indices modulo n. Unless otherwise stated, all A-modules are finite-dimensional left A-modules.

Let $Q = C_{L+1}$ be the cyclic quiver with vertex set

$$Q_0 = \{0, \dots, L\}$$

and arrows

$$a_l : l-1 \longrightarrow l, \qquad l \in \{0, \dots, L\},$$

where $a_0 : L \to 0$. We read products of arrows from right to left. For $1 \le l \le L$, define the length L path

$$p_l := a_{l-1} \cdots a_1 a_0 a_L \cdots a_{l+1}.$$

Thus p_l is the unique path of length L starting at l, ending at $l-1$, and passing through the closing arrow a_0.

Let

$$I = \langle p_1, \dots, p_L \rangle$$

and define the bound path algebra

$$A := kQ/I.$$

Equivalently, an A-module with dimension vector $d = (d_0, \dots, d_L)$ is a representation of Q on vector spaces $V_i \simeq k^{d_i}$, with linear maps

$$\phi_l : V_{l-1} \longrightarrow V_l,$$

satisfying the relations

$$\phi_{l-1} \circ \cdots \circ \phi_1 \circ \phi_0 \circ \phi_L \circ \cdots \circ \phi_{l+1} = 0, \qquad 1 \le l \le L.$$

We write $\mathrm{Rep}_d(A)$ for the corresponding representation variety.

Proposition 5.1. For $k = \mathbb{R}$, the cyclic variety Γ_d is identified with $\mathrm{Rep}_d(A)$ by

$$W_l = \phi_l, \qquad V_i \simeq \mathbb{R}^{d_i}.$$

Proof. Under this identification, the defining equations of Γ_d are exactly the relations generated by the paths $p_1, \dots, p_L$. Hence $\Gamma_d \cong \mathrm{Rep}_d(A)$. □

Let $R \subset kQ$ be the arrow ideal, i.e. the ideal generated by all arrows. Every path of length $L+1$ contains one of the paths $p_1, \dots, p_L$ as a subpath. Therefore

$$R^{L+1} \subset I.$$

It follows that A is finite-dimensional. Moreover,

$$J(A) = R/I,$$

so $J(A)$ is spanned by the nonzero paths of positive length, and

$$A/J(A) \cong \bigoplus_{i=0}^{L} k e_i.$$

In particular, the simple A-modules are

$$S_i := P_i / J P_i, \qquad P_i := A e_i.$$

Definition 5.2. A finite-length module U is called *uniserial* if its submodules are totally ordered by inclusion. Equivalently, U has a unique composition series.

A finite-dimensional algebra is called a Nakayama algebra if its indecomposable projective and indecomposable injective modules are uniserial.

Proposition 5.3. The algebra $A = kQ/I$ is a finite-dimensional Nakayama algebra. Its Kupisch series is

$$(c_0, \dots, c_L) = (L+1, L, \dots, L).$$

Proof. We first describe the indecomposable projectives. With our convention for path multiplication, $P_i = Ae_i$ is spanned by the nonzero paths starting at i. Let

$$q_{i,0} := e_i, \qquad q_{i,\ell} := a_{i+\ell} \cdots a_{i+1}$$

be the unique path of length ℓ starting at i. The path $q_{i,\ell}$ is nonzero precisely for

$$0 \le r < c_i,$$

where

$$c_0 = L+1, \qquad c_i = L \quad \text{for } 1 \le i \le L.$$

Indeed, the path starting at 0 of length L does not pass through a_0, so it is nonzero, whereas the path starting at 0 of length $L+1$ vanishes. If $i \neq 0$, then the path of length L starting at i is exactly the relation p_i, so it vanishes.

Thus

$$P_i = \langle q_{i,0}, q_{i,1}, \dots, q_{i,c_i-1} \rangle_k,$$

and

$$J^s P_i = \langle q_{i,s}, q_{i,s+1}, \dots, q_{i,c_i-1} \rangle_k, \qquad 0 \le s \le c_i.$$

In particular,

$$P_i \supset JP_i \supset J^2 P_i \supset \cdots \supset J^{c_i} P_i = 0.$$

Moreover,

$$J^s P_i / J^{s+1} P_i \cong S_{i+s}.$$

Hence this chain is a composition series.

It remains to check that it is the unique composition series. Let $N \subset P_i$ be a nonzero submodule. Since the idempotents e_v act as projections and each space $e_v P_i$ is at most one-dimensional, N is spanned by some subset of the basis elements $q_{i,\ell}$. If $q_{i,r} \in N$, then stability under the action of the arrows implies

$$q_{i,\ell+1}, q_{i,\ell+2}, \dots, q_{i,c_i-1} \in N.$$

Therefore

$$N = J^s P_i$$

for some s. Hence the submodules of P_i are totally ordered, and P_i is uniserial.

The modules $P_i = Ae_i$ form a complete set of indecomposable projective A-modules. The same argument applied to the opposite algebra shows that the indecomposable injective A-modules are also uniserial. Therefore A is a finite-dimensional Nakayama algebra. □

Lemma 5.4. Let B be a finite-dimensional Nakayama algebra. Then every indecomposable finite-dimensional B-module is uniserial.

Proof. This comes from the Artinian characterization of Nakayama algebras: for a finite-dimensional algebra B, the following are equivalent:

1. B is Nakayama;
2. every indecomposable finite-dimensional B-module is uniserial;
3. every finite-dimensional B-module is a direct sum of uniserial modules.

Since B is finite-dimensional, it is Artinian, so this characterization applies. □

Definition 5.5. For $i, j \in Q_0$, let

$$\delta(i,j) \in \{0, \dots, L\}$$

be the unique integer such that

$$j \equiv i + \delta(i,j) \pmod{L+1}.$$

Whenever

$$\delta(i,j) + 1 \le c_i,$$

define the interval module $M(i,j)$ by

$$M(i,j)_v = \begin{cases} k, & \text{if } v \equiv i + \ell \pmod{L+1} \text{ for all } 0 \le \ell \le \delta(i,j), \\ 0, & \text{otherwise,} \end{cases}$$

and

$$M(i,j)_{a_m} = \begin{cases} \mathrm{id}_k, & \text{if } m \equiv i + \ell \pmod{L+1} \text{ for all } 1 \le \ell \le \delta(i,j), \\ 0, & \text{otherwise.} \end{cases}$$

The condition $\delta(i,j) + 1 \le c_i$ is exactly the condition that the relations in I are satisfied.

Theorem 5.6. Every nonzero indecomposable finite-dimensional A-module is isomorphic to a unique interval module $M(i,j)$ with

$$\delta(i,j) + 1 \le c_i.$$

Equivalently, the indecomposable A-modules are given by:

$$M(i, i+\ell-1), \qquad 0 \le i \le L, \quad 1 \le \ell \le c_i,$$

where indices are taken modulo $L+1$.

Proof. Let M be a nonzero indecomposable finite-dimensional A-module. Since A is a finite-dimensional Nakayama algebra, the previous lemma implies that M is uniserial. Hence M has a unique maximal submodule, and therefore its top is simple:

$$\operatorname{top} M := M/JM \cong S_i$$

for a unique vertex i.

Let

$$\pi : P_i \twoheadrightarrow M$$

be the projective cover of M. Such a projective cover exists because A is finite-dimensional, and it must be P_i because $\operatorname{top} P_i \cong S_i$. Let $K = \ker \pi$. Then

$$M \cong P_i/K.$$

Since P_i is uniserial, its submodules are precisely

$$P_i,\ JP_i,\ J^2P_i, \dots,\ J^{c_i}P_i = 0.$$

Because $M \neq 0$, the kernel is not all of P_i. Thus there is a unique integer ℓ with $1 \leq \ell \leq c_i$ such that

$$K = J^\ell P_i.$$

Therefore

$$M \cong P_i/J^\ell P_i.$$

As a representation of the quiver, $P_i/J^l P_i$ has one-dimensional vector spaces at the vertices

$$i,\ i+1,\ \dots,\ i+\ell-1$$

and zero vector spaces elsewhere. The arrow maps along this interval are id_k, and all other arrow maps are zero. Hence

$$P_i/J^\ell P_i \cong M(i, i+\ell-1).$$

Conversely, for every i and every $1 \leq r \leq c_i$, the interval module $M(i, i+\ell-1)$ is isomorphic to $P_i/J^\ell P_i$. Since it is a nonzero quotient of a uniserial module, it is uniserial, hence indecomposable.

Finally, uniqueness follows because the top of $M(i, i+\ell-1)$ is S_i, which determines i, and its composition length is ℓ, which determines the endpoint $i+\ell-1$. This proves the classification. □

Definition 5.7. Let

$$c_0 = L+1, \qquad c_i = L \quad \text{for } 1 \leq i \leq L.$$

Define the set of admissible intervals by

$$\mathcal{I} := \{(i,\ell) \mid i \in \{0, \dots, L\},\ 1 \leq \ell \leq c_i\}.$$

For $(i,\ell) \in \mathcal{I}$, write

$$j(i,\ell) := i+\ell-1 \pmod{L+1}.$$

The corresponding interval module is

$$M_\ell(i) := M(i, j(i,\ell)).$$

Equivalently, $M_\ell(i)$ is supported on the nonzero path

$$i \to i+1 \to \cdots \to i+\ell-1,$$

with one-dimensional vector spaces along this path and identity maps along the arrows of the path.

The number of indecomposables is

$$|\mathcal{I}| = c_0 + \sum_{i=1}^{L} c_i = (L+1) + L^2 = L^2 + L + 1.$$

Corollary 5.8 (Krull–Schmidt decomposition for $\mathrm{Rep}_d(A)$)**.** Every finite-dimensional A-module M admits a decomposition

$$M \cong \bigoplus_{(i,\ell)\in\mathcal{I}} M_\ell(i)^{\oplus m_{i,\ell}},$$

where $m_{i,\ell} \in \mathbb{Z}_{\geq 0}$, and only finitely many multiplicities are nonzero.

Moreover, the multiplicities $m_{i,\ell}$ are uniquely determined by M. Equivalently, the decomposition is unique up to permutation and isomorphism of the indecomposable summands.

Proof. Since A is finite-dimensional, the category of finite-dimensional A-modules is a Krull–Schmidt category. Hence every finite-dimensional A-module decomposes as a finite direct sum of indecomposable A-modules, and this decomposition is unique up to permutation and isomorphism of the indecomposable summands.

By the classification theorem, every nonzero indecomposable A-module is isomorphic to an interval module

$$M_\ell(i) \cong P_i/J^\ell P_i$$

for a unique pair

$$(i, \ell) \in \mathcal{I}.$$

Therefore every finite-dimensional A-module has a decomposition of the form

$$M \cong \bigoplus_{(i,\ell)\in\mathcal{I}} M_\ell(i)^{\oplus m_{i,\ell}}.$$

The uniqueness statement in Krull–Schmidt implies that the multiplicities $m_{i,\ell}$ are uniquely determined. □

5.3 From multiplicity patterns to rank patterns

Definition 5.9 (Multiplicity Patterns)**.** Consider a A-module M of vector dimension d and its Krull-Schmidt decomposition into $M_\ell(i)$ for $(i, \ell) \in \mathcal{I}$. For each vertex k the dimension d_k of the representation M at vertex k satisfies the dimension constraint:

$$d_k \;=\; \sum_{(i,\ell)\in\mathcal{I}} m_{i,\ell}\,\mathbf{1}\{k \in \{i, i+1, \ldots, i+\ell\} \pmod{L+1}\} \quad k = 0, \ldots, L$$

Define the set of *multiplicity patterns* as

$$\mathcal{M}_d^+ := \left\{ (m_{i,\ell}) \in \mathbb{Z}_{\geq 0}^{\mathcal{I}} \;\middle|\; d_k \;=\; \sum_{(i,\ell)\in\mathcal{I}} m_{i,\ell}\,\mathbf{1}\{k \in \{i, i+1, \ldots, i+\ell\} \pmod{L+1}\} \quad k = 0, \ldots, L \right\}.$$

We want to stratify the critical loci in terms of the rank patterns as they are more intuitive and native to the DLN representation. In this section we show that there is a bijection between the multiplicity patterns and a set of cyclic rank patterns satisfying some positivity constraint.

Definition 5.10. For $(a, b) \in \mathcal{I}$, let

$$q_{ab}$$

denote the unique nonzero oriented path of A over the interval $[a, b]$ from vertex a to vertex b.

Definition 5.11 (Cyclic rank pattern)**.** Let M be an A-module of dimension vector d. Its lifted cyclic rank pattern is the array

$$\rho(M) = (\rho_{ab}(M))_{(a,b)\in\mathcal{I}}, \qquad \rho_{ab}(M) := \operatorname{rank}\big(q_{ab}(M)\big).$$

Two coordinates play a distinguished role:

$$\rho_{0L}(M) = \operatorname{rank}(q_{0L}(M)), \qquad \rho_{L,0}(M) = \operatorname{rank}(q_{L,0}(M)).$$

In the DLN representation these are exactly the end-to-end rank and the rank of the closing arrow.

Proposition 5.12 (Forward formula)**.** Let M have multiplicity pattern

$$m = (m_{uv})_{(u,v)\in\mathcal{I}}.$$

Then for every $(a,b) \in \mathcal{I}_L^{\mathrm{wt}}$,

$$\rho_{ab}(M) = \sum_{\substack{(u,v)\in\mathcal{I} \\ u\le a\le b\le v}} m_{uv}.$$

Proof. On an indecomposable interval module $M(u,v)$, the path q_{ab} acts with rank 1 exactly when the path interval $[a,b]$ is contained in $[u,v]$, and otherwise with rank 0. The claim follows by additivity of rank on direct sums. □

Proposition 5.13 (Inversion formula)**.** Extend every rank pattern by the convention

$$\rho_{uv} = 0 \qquad \text{for } (u,v) \notin \mathcal{I}.$$

Then for every admissible interval $(a,b) \in \mathcal{I}$,

$$m_{ab} = \rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1}.$$

Proof. Insert the forward formula into the right-hand side and note that, for every fixed interval (u,v), the coefficient of m_{uv} is

$$\mathbf{1}_{u\le a\le b\le v} - \mathbf{1}_{u\le a-1\le b\le v} - \mathbf{1}_{u\le a\le b+1\le v} + \mathbf{1}_{u\le a-1\le b+1\le v}.$$

This coefficient equals 1 precisely when $(u,v) = (a,b)$ and equals 0 otherwise. A more detailed telescoping proof is recorded in Appendix 8.3. □

Definition 5.14 (Admissible rank-pattern set)**.**

$$\mathcal{R}_d := \left\{ \rho \in \mathbb{Z}_{\ge 0}^{\mathcal{I}} \;\middle|\; \begin{array}{r} \rho_{aa} = d_a \text{ for every } 0 \le a \le L, \\ \rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1} \ge 0 \\ \text{for every } (a,b) \in \mathcal{I} \end{array} \right\}.$$

Proposition 5.15 (Multiplicity-rank bijection)**.** The multiplicity-pattern set and the rank-pattern set are canonically in bijection:

$$\mathcal{M}_d^+ \xrightarrow{\sim} \mathcal{R}_d.$$

Proof. The forward map sends a multiplicity pattern to the rank pattern given by Proposition 5.12. The inverse map is the finite-difference formula of Proposition 5.13. The diagonal condition $\rho_{aa} = d_a$ ensures that the resulting multiplicity pattern has dimension vector d. □

6 Stratification of the critical loci with cyclic rank patterns

We have identified the cyclic variety Γ_d with the representation variety of the Nakayama algebra $A = kQ/I$ with dimension vector d. Under this identification, the G_d-orbits in Γ_d are precisely the isomorphism classes of A-modules with dimension vector d. By the Krull–Schmidt theorem, each such module has a unique decomposition into indecomposable Nakayama modules, and hence each orbit is indexed by a multiplicity pattern $m \in \mathcal{M}_d^+$.

Proposition 6.1 (Orbit stratification of Γ_d by indecomposable Nakayama modules)**.** Fix $L \geq 1$, let $Q = C_{L+1}$ be the oriented cyclic quiver on vertices $\{0, \ldots, L\}$, and let $A = kQ/I$ be the Nakayama quiver algebra defined above. Then

$$\Gamma_d = \bigsqcup_{m \in \mathcal{M}_d^+} \mathcal{O}_m = \bigsqcup_{m \in \mathcal{M}_d^+} G_d \cdot \phi_m,$$

where ϕ_m is any representative whose associated A-module has multiplicity pattern m. In particular, the G_d-orbits in Γ_d are parametrized by the multiplicity data of indecomposable A-modules.

Using the multiplicity-rank bijection of Proposition 5.15, we may equivalently index the same orbits by cyclic rank patterns. Thus we write

$$\Gamma_d = \bigsqcup_{\rho \in \mathcal{R}_d} \mathcal{O}_\rho,$$

where $\mathcal{O}_\rho$ is the G_d-orbit with cyclic rank pattern ρ. We use the coordinatewise rank order

$$\rho' \leq_{\mathrm{rk}} \rho \quad \Longleftrightarrow \quad \rho'_{ab} \leq \rho_{ab} \quad \text{for every cyclic rank coordinate } (a,b) \in \mathcal{I}.$$

The orbit stratification of the cyclic variety satisfies the closure formula

$$\overline{\mathcal{O}_\rho}^{\Gamma_d} = \bigsqcup_{\substack{\rho' \in \mathcal{R}_d \\ \rho' \leq_{\mathrm{rk}} \rho}} \mathcal{O}_{\rho'}.$$

Equivalently, the frontier of $\mathcal{O}_\rho$ in Γ_d is the union of the strictly lower coordinatewise rank-pattern orbits.

We now specialize this stratification to a fixed S-critical locus. Put

$$B_S := P_S W^\star, \qquad E_S := \Sigma_{XY} P_Q,$$

and recall that

$$\Gamma_d^S := \pi^{-1}(B_S, E_S) \subseteq \Gamma_d.$$

By Proposition 4.1, the map

$$\Theta_S : \mathcal{C}_d^S \longrightarrow \Gamma_d^S, \qquad \Theta_S(\theta) = (\theta, E_S),$$

identifies the S-critical locus with the fiber Γ_d^S.

Let

$$r := \mathrm{rank}(B_S), \qquad q_S := \mathrm{rank}(E_S).$$

Under the nondegeneracy assumptions of the main theorem, $r = |S|$ and $q_S = d_L - r$. The two distinguished cyclic rank coordinates satisfy

$$\rho_{0L} = \mathrm{rank}(\mu(\theta)), \qquad \rho_{L0} = \mathrm{rank}(W_0).$$

Hence every point of Γ_d^S has

$$\rho_{0L} = r, \qquad \rho_{L0} = q_S.$$

These rank conditions are necessary but not sufficient for a cyclic orbit to meet the fixed fiber Γ_d^S.

For each $\rho \in \mathcal{R}_d$, define the induced fixed-S stratum

$$\mathcal{O}_\rho^S := \mathcal{O}_\rho \cap \Gamma_d^S = \mathcal{O}_\rho \cap \pi^{-1}(B_S, E_S).$$

We keep only the nonempty intersections and define

$$\mathcal{R}_d^S := \left\{\rho \in \mathcal{R}_d \;\middle|\; \rho_{0L} = r,\ \rho_{L0} = q_S,\ \mathcal{O}_\rho \cap \Gamma_d^S \neq \varnothing\right\}.$$

Thus

$$\Gamma_d^S = \bigsqcup_{\rho \in \mathcal{R}_d^S} \mathcal{O}_\rho^S.$$

The relevant symmetry group for the fixed fiber is the stabilizer of the pair (B_S, E_S). Define

$$K_S := \left\{g \in G_d \;\middle|\; g_L B_S g_0^{-1} = B_S,\ g_0 E_S g_L^{-1} = E_S\right\}.$$

The equivariance of π,

$$\pi(g \cdot (\theta, W_0)) = (g_L \mu(\theta) g_0^{-1},\, g_0 W_0 g_L^{-1}),$$

shows that K_S acts on Γ_d^S.

Theorem 6.2 (Induced orbit stratification of the S-critical locus)**.** For every $\rho \in \mathcal{R}_d^S$, the stratum

$$\mathcal{O}_\rho^S = \mathcal{O}_\rho \cap \Gamma_d^S$$

is a single K_S-orbit. Consequently, from proposition 4.2

$$\Gamma_d^S = \bigsqcup_{\rho \in \mathcal{R}_d^S} \mathcal{O}_\rho^S$$

is a finite stratification into smooth locally closed K_S-orbits. Define the fixed-S closure relation by

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \subseteq \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S}.$$

Equivalently, since the strata are K_S-orbits,

$$\rho' \leq_S \rho \quad \Longleftrightarrow \quad \mathcal{O}_{\rho'}^S \cap \overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \neq \varnothing.$$

Then

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} = \bigsqcup_{\substack{\rho' \in \mathcal{R}_d^S \\ \rho' \leq_S \rho}} \mathcal{O}_{\rho'}^S.$$

Equivalently, if

$$\rho' <_S \rho \quad \Longleftrightarrow \quad \rho' \leq_S \rho \text{ and } \rho' \neq \rho,$$

then the frontier satisfies

$$\overline{\mathcal{O}_\rho^S}^{\Gamma_d^S} \setminus \mathcal{O}_\rho^S = \bigsqcup_{\substack{\rho' \in \mathcal{R}_d^S \\ \rho' <_S \rho}} \mathcal{O}_{\rho'}^S.$$

Moreover, the fixed-S closure order refines the ambient coordinatewise rank order:

$$\rho' \leq_S \rho \quad \Longrightarrow \quad \rho' \leq_{\mathrm{rk}} \rho.$$

Thus

$$\{\rho' \in \mathcal{R}_d^S \mid \rho' \leq_S \rho\} \subseteq \{\rho' \in \mathcal{R}_d^S \mid \rho' \leq_{\mathrm{rk}} \rho\}.$$

The inclusion may be strict. Via the identification $\mathcal{C}_d^S \cong \Gamma_d^S$, define

$$\mathcal{C}_\rho^S := \Theta_S^{-1}(\mathcal{O}_\rho^S).$$

Then

$$\mathcal{C}_d^S = \bigsqcup_{\rho \in \mathcal{R}_d^S} \mathcal{C}_\rho^S.$$

Consequently, we have a stratification of the variety of the critical points:

$$\mathcal{C}_d = \bigsqcup_r \mathcal{C}_d^r = \bigsqcup_r \bigsqcup_{|S|=r} \mathcal{C}_d^S = \bigsqcup_r \bigsqcup_{|S|=r} \bigsqcup_{\rho \in \mathcal{R}_d^S} \mathcal{C}_\rho^S.$$

Equivalently, after identifying $\mathcal{C}_d^S$ with Γ_d^S, the last strata may be denoted by the corresponding K_S-orbits $\mathcal{O}_\rho^S$.

Remark 6.3 (Compatibility of the fixed product fiber and choice of base field)**.** Set

$$B_S := P_S W^*, \qquad E_S := \Sigma_{XY} P_Q.$$

Then

$$E_S B_S = \Sigma_{XY} P_Q P_S W^* = 0$$

because $P_Q P_S = 0$. However, the other composition

$$B_S E_S = P_S W^* \Sigma_{XY} P_Q$$

does not vanish in general. It is an additional compatibility condition.

This compatibility is necessary for the fixed product fiber

$$\pi^{-1}(B_S, E_S) \subseteq \Gamma_d$$

to be nonempty. Indeed, if

$$(\theta, W_0) \in \Gamma_d, \qquad \mu(\theta) = W_L \cdots W_1,$$

then the cyclic relations imply

$$\mu(\theta) W_0 = 0, \qquad W_0 \mu(\theta) = 0.$$

For example, the relation with $\ell = L$ gives

$$W_{L-1} \cdots W_1 W_0 = 0,$$

and hence

$$\mu(\theta) W_0 = W_L (W_{L-1} \cdots W_1 W_0) = 0.$$

Similarly, the relation with $\ell = 1$ gives

$$W_0 W_L \cdots W_2 = 0,$$

and hence

$$W_0 \mu(\theta) = (W_0 W_L \cdots W_2) W_1 = 0.$$

Therefore, if

$$\pi(\theta, W_0) = (B_S, E_S),$$

then necessarily

$$B_S E_S = 0, \qquad E_S B_S = 0.$$

Consequently, if $B_S E_S \neq 0$, then

$$\pi^{-1}(B_S, E_S) = \varnothing,$$

and the corresponding S-critical locus is empty.

In our deep linear network setting, this condition is satisfied when P_S and P_Q are the spectral projectors associated with the left singular directions of the teacher matrix

$$M := \Sigma_{YX} \Sigma_X^{-1/2}.$$

Indeed,

$$W^* \Sigma_{XY} = \Sigma_{YX} \Sigma_X^{-1} \Sigma_{XY} = M M^\top.$$

Hence, if P_S and P_Q are complementary spectral projectors for $MM^\top$, then

$$B_S E_S = P_S W^* \Sigma_{XY} P_Q = P_S M M^\top P_Q = 0.$$

If instead P_S is defined using the left singular vectors of W^* itself, then $B_S E_S = 0$ need not hold unless, for example, $\Sigma_X = I$.

We now discuss the base field. The representation-theoretic constructions are valid over both $k = \mathbb{R}$ and $k = \mathbb{C}$. The cyclic variety $\Gamma_{d,k}$, the group $G_{d,k}$, and the orbit strata $\mathcal{O}_{\rho,k}$ are defined over k. The orbit decomposition is field-independent: the indecomposable Nakayama modules have representatives with matrices whose entries lie in $\{0, 1\}$. Thus the same cyclic rank patterns occur over $\mathbb{R}$ and over $\mathbb{C}$, and the k-rational points of each nonempty orbit are Zariski dense in that orbit.

Assume now that the fixed pair (B_S, E_S) is compatible, i.e.

$$B_S E_S = 0, \qquad E_S B_S = 0.$$

Let

$$r := \operatorname{rank}(B_S), \qquad q_S := \operatorname{rank}(E_S).$$

The pair (B_S, E_S) is then a representation of the two-vertex radical-square-zero cyclic quiver

$$V_0 \rightleftarrows V_L, \qquad BE = 0, \quad EB = 0.$$

Such pairs are classified, over either $\mathbb{R}$ or $\mathbb{C}$, by the two ranks r and q_S. Equivalently, after choosing bases, the pair is conjugate to the normal form

$$B_{r,q_S}(u, v, w) = (u, 0, 0), \qquad E_{r,q_S}(a, b, c) = (0, b, 0),$$

with respect to decompositions

$$k^{d_0} \cong k^r \oplus k^{q_S} \oplus k^{d_0 - r - q_S}, \qquad k^{d_L} \cong k^r \oplus k^{q_S} \oplus k^{d_L - r - q_S},$$

whenever these dimensions are nonnegative. This normal form is defined over $\{0, 1\}$, so the corresponding real and complex orbits have the same algebraic geometry.

Therefore, for the fixed fiber

$$\Gamma_{d,k}^S := \pi_k^{-1}(B_S, E_S),$$

the induced strata

$$\mathcal{O}^S_{\rho,k} := \mathcal{O}_{\rho,k} \cap \Gamma^S_{d,k}$$

are independent of the choice $k = \mathbb{R}$ or $k = \mathbb{C}$, provided all objects are taken over the same base field. More precisely,

$$\mathcal{O}^S_{\rho,k} \neq \varnothing$$

if and only if the outer cyclic rank coordinates satisfy

$$\rho_{0L} = r, \qquad \rho_{L0} = q_S.$$

Equivalently,

$$\mathcal{R}^S_d = \{\rho \in \mathcal{R}_d \mid \rho_{0L} = r,\ \rho_{L0} = q_S\}.$$

If one does not want to assume the compatibility condition explicitly, the safe definition is instead

$$\mathcal{R}^S_d := \{\rho \in \mathcal{R}_d \mid \mathcal{O}_\rho \cap \pi^{-1}(B_S, E_S) \neq \varnothing\}.$$

For every nonempty induced stratum, the stabilizer

$$K_S = \left\{g \in G_d \;\middle|\; g_L B_S g_0^{-1} = B_S,\ g_0 E_S g_L^{-1} = E_S\right\}$$

acts transitively:

$$\mathcal{O}^S_\rho = K_S \cdot x$$

for any $x \in \mathcal{O}^S_\rho$. Hence the K_S-orbit stratification and its Zariski frontier relation are the same over $\mathbb{R}$ and $\mathbb{C}$, after complexification.

The only distinction we do not identify across base fields is the ordinary real analytic connected-component decomposition. A real locus of a complex algebraic orbit may have more than one Euclidean connected component. This does not affect the cyclic rank-pattern labels, the K_S-orbit stratification, the Zariski closure relation, or the codimension computations.

7 Discussion

Have shown that the critical points can be stratified into strata parameterized by geometric invariant: cyclic rank patterns. This stratification provides a combinatorial description of the loss landscape of DLNs. This is useful as a coarse grained description of the loss landscape. The loss landscape can be thought as a graph that is hierachically structured with particular nodes corresponding to a strata given by the rank r, the subset S of singular vectors and cyclic rank patterns $\rho \in \mathcal{R}^S_d$. The stratification also induce some constraint on the training dynamics. The strata with high rank patterns are less degenerate and over a given S-critical loci the learning process can only move toward more degenerate strata which generalise the results from [**?**]. We also discover geometrically principled observables: rank patterns. Rank patterns are useful observables to discuss the implicit biases of the learning process, for example SGD. Indeed using the rank patterns we can make explicit an implicit bias of SGD during training which over a given critical loci will favour solution with certain rank patterns. For example in the case of DLN at small initialization we expect from [**?**] that SGD will have an implicit biases toward more degenerate strata i.e. solutions with low rank patterns. We should be able to make these implicit biases explicit by showing that SGD reduce some free energy functional of the student W with an entropic regularization depending on the cyclic rank patterns or some corresponding continuous version using the singular values of partial product of weight matrices of the DLN.

7.1 Future work

Promising next step would be to derive a codimension formula extending the results of [**?**] to all the local minima and the saddle points. One could also relax the assumptions of non degenerate data and invertible input covariance matrix. Future work could also make the implicit biases of SGD more explicit using the geometric invariant and use the stratification to coarse grain the training dynamics of SGD over the loss landscape. For example one could model SGD as a random walk over a graph with nodes indexed by the strata and edges corresponding to transition between strata. Another exciting direction would be investigate the quiver representation theoretic perspective on non-linear models for example ReLU networks to derive useful geometric invariant and a combinatorial description of ReLU loss landscapes.

8 Appendix

8.1 Detailed proof of Proposition 4.1

We prove the equality

$$\mathcal{C}_d^S = \Gamma_d^S \cap \mu^{-1}(P_S W^*) = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*)$$

as an equality of subsets of Ω.

Recall that

$$\iota_S(\theta) = (\theta, \Sigma_{XY} P_Q),$$

and therefore

$$\Gamma_d^S = \iota_S^{-1}(\Gamma_d^r).$$

Thus $\theta \in \Gamma_d^S$ means that

$$(\theta, \Sigma_{XY} P_Q) \in \Gamma_d^r.$$

Equivalently,

$$\operatorname{rank}(\mu(\theta)) = r$$

and

$$W_{<\ell} \Sigma_{XY} P_Q W_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

We first prove

$$\mathcal{C}_d^S \subseteq \Gamma_d^S \cap \mu^{-1}(P_S W^*).$$

Let $\theta \in \mathcal{C}_d^S$. By definition of the S-critical locus,

$$\nabla_\theta \mathcal{L}(\theta) = 0 \qquad \text{and} \qquad \mu(\theta) = P_S W^*.$$

Since $\operatorname{rank}(P_S W^*) = r$, it follows that

$$\operatorname{rank}(\mu(\theta)) = r.$$

It remains to show the cyclic equations with closing arrow $\Sigma_{XY} P_Q$.

For each layer ℓ, the transpose-gradient formula is

$$\big(\nabla_{W_\ell} \mathcal{L}(\theta)\big)^\top = 2 W_{<\ell} \big(\Sigma_X \mu(\theta)^\top - \Sigma_{XY}\big) W_{>\ell}.$$

Using $\mu(\theta) = P_S W^*$, we get

$$\Sigma_X \mu(\theta)^\top - \Sigma_{XY} = \Sigma_X (P_S W^*)^\top - \Sigma_{XY}.$$

Since P_S is an orthogonal projector, $P_S^\top = P_S$, and therefore

$$(P_S W^*)^\top = (W^*)^\top P_S.$$

Moreover, by the definition

$$W^* = \Sigma_{YX}\Sigma_X^{-1},$$

and by symmetry of Σ_X, we have

$$\Sigma_X (W^*)^\top = \Sigma_X (\Sigma_X^{-1})^\top \Sigma_{YX}^\top = \Sigma_X \Sigma_X^{-1} \Sigma_{XY} = \Sigma_{XY}.$$

Hence

$$\Sigma_X (P_S W^*)^\top - \Sigma_{XY} = \Sigma_X (W^*)^\top P_S - \Sigma_{XY} = \Sigma_{XY} P_S - \Sigma_{XY}.$$

Since $P_Q = I - P_S$, this becomes

$$\Sigma_X (P_S W^*)^\top - \Sigma_{XY} = -\Sigma_{XY} P_Q.$$

Substituting into the gradient formula gives

$$\big(\nabla_{W_\ell} \mathcal{L}(\theta)\big)^\top = -2 W_{<\ell} \Sigma_{XY} P_Q W_{>\ell}.$$

Because $\theta \in \mathcal{C}_d^S$, every partial gradient vanishes. Therefore

$$W_{<\ell} \Sigma_{XY} P_Q W_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

Thus

$$(\theta, \Sigma_{XY} P_Q) \in \Gamma_d^r,$$

or equivalently,

$$\theta \in \Gamma_d^S.$$

Since also $\mu(\theta) = P_S W^*$, we have

$$\theta \in \Gamma_d^S \cap \mu^{-1}(P_S W^*).$$

Conversely, suppose that

$$\theta \in \Gamma_d^S \cap \mu^{-1}(P_S W^*).$$

Then

$$\mu(\theta) = P_S W^*$$

and, because $\theta \in \Gamma_d^S$,

$$W_{<\ell} \Sigma_{XY} P_Q W_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

As above,

$$\Sigma_X \mu(\theta)^\top - \Sigma_{XY} = -\Sigma_{XY} P_Q.$$

Therefore, for every ℓ,

$$\big(\nabla_{W_\ell} \mathcal{L}(\theta)\big)^\top = -2 W_{<\ell} \Sigma_{XY} P_Q W_{>\ell} = 0.$$

Hence

$$\nabla_{W_\ell} \mathcal{L}(\theta) = 0 \qquad \text{for every } 1 \le \ell \le L.$$

So

$$\nabla_\theta \mathcal{L}(\theta) = 0.$$

Together with $\mu(\theta) = P_S W^*$, this gives

$$\theta \in \mathcal{C}_d^S.$$

This proves the reverse inclusion and therefore the proposition.

8.2 Detailed proof of Proposition 6.2

Assume that

$$\Gamma_d^r = \bigsqcup_{\rho \in R} O_\rho^r$$

is a finite decomposition into locally closed subsets of Γ_d^r, and that the pieces satisfy

$$\overline{O_\rho^r}^{\Gamma_d^r} \subseteq \bigcup_{\rho' \leq \rho} O_{\rho'}^r.$$

For each $\rho \in R$, define

$$\mathcal{C}_\rho^S := \iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*).$$

We first prove that the pieces $\mathcal{C}_\rho^S$ give a disjoint decomposition of $\mathcal{C}_d^S$. By Proposition 4.1,

$$\mathcal{C}_d^S = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*).$$

Since

$$\Gamma_d^r = \bigsqcup_{\rho \in R} O_\rho^r,$$

we have

$$\iota_S^{-1}(\Gamma_d^r) = \iota_S^{-1}\left(\bigsqcup_{\rho \in R} O_\rho^r\right).$$

Preimages commute with arbitrary unions, and preimages of disjoint sets are disjoint. Hence

$$\iota_S^{-1}(\Gamma_d^r) = \bigsqcup_{\rho \in R} \iota_S^{-1}(O_\rho^r).$$

Intersecting both sides with $\mu^{-1}(P_S W^*)$ gives

$$\mathcal{C}_d^S = \left(\bigsqcup_{\rho \in R} \iota_S^{-1}(O_\rho^r)\right) \cap \mu^{-1}(P_S W^*).$$

Therefore

$$\mathcal{C}_d^S = \bigsqcup_{\rho \in R} \left(\iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*)\right) = \bigsqcup_{\rho \in R} \mathcal{C}_\rho^S.$$

After discarding the empty terms, this is the claimed decomposition.

Next we prove local closedness. Each O_ρ^r is locally closed in Γ_d^r. Since Γ_d^r is a locally closed subset of $\Omega \times \mathrm{Mat}_{d_0 \times d_L}$, each O_ρ^r is locally closed in $\Omega \times \mathrm{Mat}_{d_0 \times d_L}$. The map

$$\iota_S : \Omega \longrightarrow \Omega \times \mathrm{Mat}_{d_0 \times d_L}$$

is continuous, so

$$\iota_S^{-1}(O_\rho^r)$$

is locally closed in Ω. The map $\mu : \Omega \to \mathrm{Mat}_{d_L \times d_0}$ is regular, hence continuous, and the point $\{P_S W^*\}$ is closed. Therefore

$$\mu^{-1}(P_S W^*)$$

is closed in Ω. Consequently

$$\mathcal{C}_\rho^S = \iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*)$$

is locally closed in Ω. Since

$$\mathcal{C}_\rho^S \subseteq \mathcal{C}_d^S,$$

it is also locally closed in $\mathcal{C}_d^S$ with the subspace topology.

It remains to prove the frontier inclusion

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

We use the following elementary fact. If $f : X \to Y$ is continuous and $A \subseteq Y$, then

$$\overline{f^{-1}(A)}^X \subseteq f^{-1}\left(\overline{A}^Y\right).$$

Indeed, $f^{-1}(\overline{A}^Y)$ is closed in X and contains $f^{-1}(A)$, so it contains the closure of $f^{-1}(A)$.

Apply this fact to the continuous map

$$\iota_S|_{\mathcal{C}_d^S} : \mathcal{C}_d^S \longrightarrow \Gamma_d^r.$$

This restriction is well-defined because, by Proposition 4.1,

$$\mathcal{C}_d^S \subseteq \Gamma_d^S = \iota_S^{-1}(\Gamma_d^r),$$

so $\iota_S(\theta) \in \Gamma_d^r$ for every $\theta \in \mathcal{C}_d^S$.

Now observe that

$$\mathcal{C}_\rho^S = (\iota_S|_{\mathcal{C}_d^S})^{-1}(O_\rho^r).$$

Therefore

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} = \overline{(\iota_S|_{\mathcal{C}_d^S})^{-1}(O_\rho^r)}^{\mathcal{C}_d^S} \subseteq (\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\overline{O_\rho^r}^{\Gamma_d^r}\right).$$

Using the assumed frontier inclusion,

$$\overline{O_\rho^r}^{\Gamma_d^r} \subseteq \bigcup_{\rho' \leq \rho} O_{\rho'}^r,$$

we obtain

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq (\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\bigcup_{\rho' \leq \rho} O_{\rho'}^r\right).$$

Preimages commute with unions, so

$$(\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\bigcup_{\rho' \leq \rho} O_{\rho'}^r\right) = \bigcup_{\rho' \leq \rho} (\iota_S|_{\mathcal{C}_d^S})^{-1}(O_{\rho'}^r).$$

But

$$(\iota_S|_{\mathcal{C}_d^S})^{-1}(O_{\rho'}^r) = \mathcal{C}_{\rho'}^S.$$

Hence

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

This proves the proposition.

The argument gives an inclusion rather than an equality. The possible failure of equality comes from the strictness of

$$\overline{f^{-1}(A)}^{X} \subseteq f^{-1}\left(\overline{A}^{Y}\right)$$

for a general continuous map f. In the present setting, a point in $\iota_S^{-1}(\overline{O_\rho^r}^{\Gamma_d^r})$ may be reachable as a degeneration of O_ρ^r in the full cyclic variety while not being reachable by a degeneration that remains inside the fixed specialization determined by ι_S. Therefore the natural statement for the S-critical locus is the one-sided frontier condition above.

8.3 Proof of the Inversion Formula

We give a slightly more explicit version of the proof of Proposition 5.13. For every admissible interval (a, b), the forward formula gives

$$\rho_{ab} = \sum_{u \le a \le b \le v} m_{uv}.$$

Therefore

$$\rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1} = \sum_{u,v} m_{uv}\, \Xi_{uv},$$

where

$$\Xi_{uv} := \mathbf{1}_{u \le a \le b \le v} - \mathbf{1}_{u \le a-1 \le b \le v} - \mathbf{1}_{u \le a \le b+1 \le v} + \mathbf{1}_{u \le a-1 \le b+1 \le v}.$$

If $u < a$, then the first two indicators cancel, and likewise the last two indicators cancel. If $v > b$, then the first and third indicators cancel, and likewise the second and fourth indicators cancel. The only way for a nonzero coefficient to remain is therefore

$$u = a, \qquad v = b.$$

In that case

$$\Xi_{ab} = 1.$$

Thus the entire sum reduces to m_{ab}, proving the inversion formula.

9 Deprecated